Power



File: power.kicad_sch

Vehicle (OBD, CAN, K-Line)



File: vehicle.kicad_sch

Backup battery



File: battery.kicad_sch

MCU (ESP32-C5, USB)



File: mcu.kicad_sch

LTE + GNSS (EC200U)



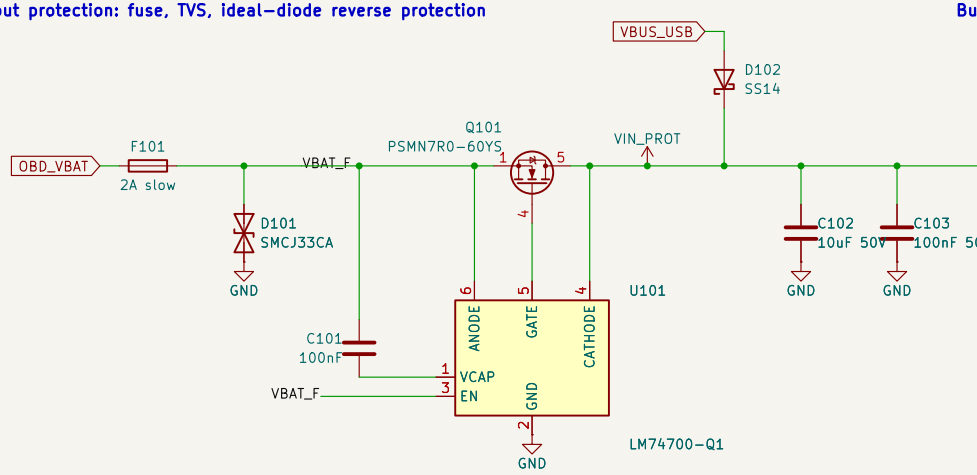
File: lte.kicad_sch

Sheet: /
File: obd-v2x-dongle.kicad_sch

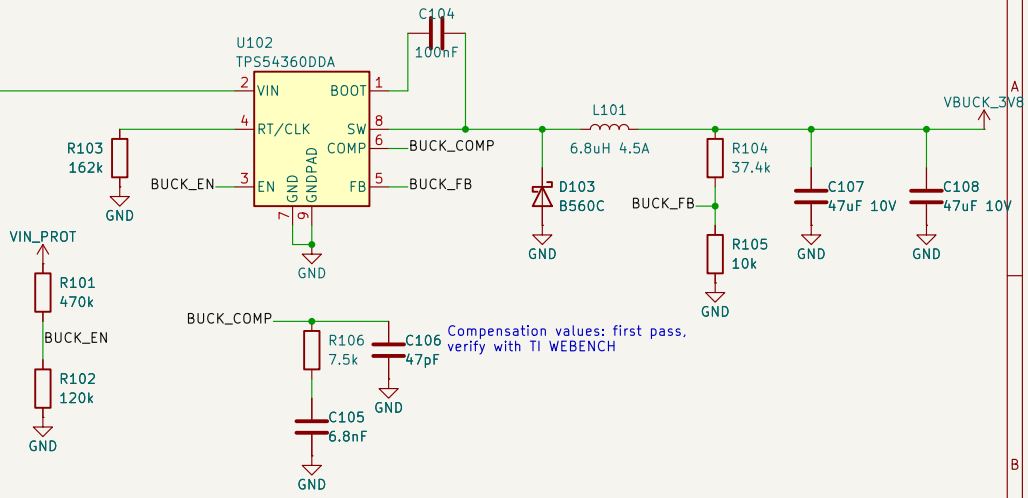
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/6

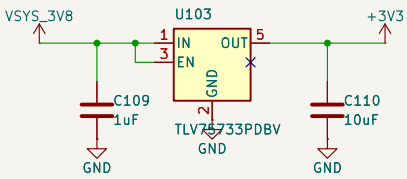
Input protection: fuse, TVS, ideal-diode reverse protection



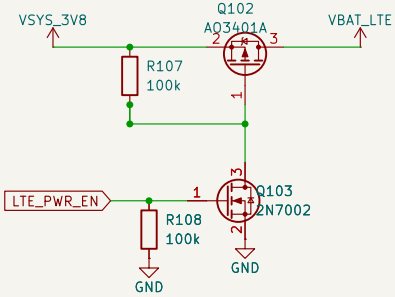
Buck 3.8 V / 3.5 A (TPS54360, 600 kHz)



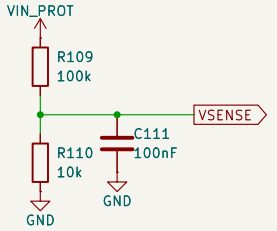
LDO 3.3 V (logic)



LTE modem supply switch (LTE_PWR_EN high = on)



Battery sense (VSENSE = VIN / 11, full scale ~35 V)



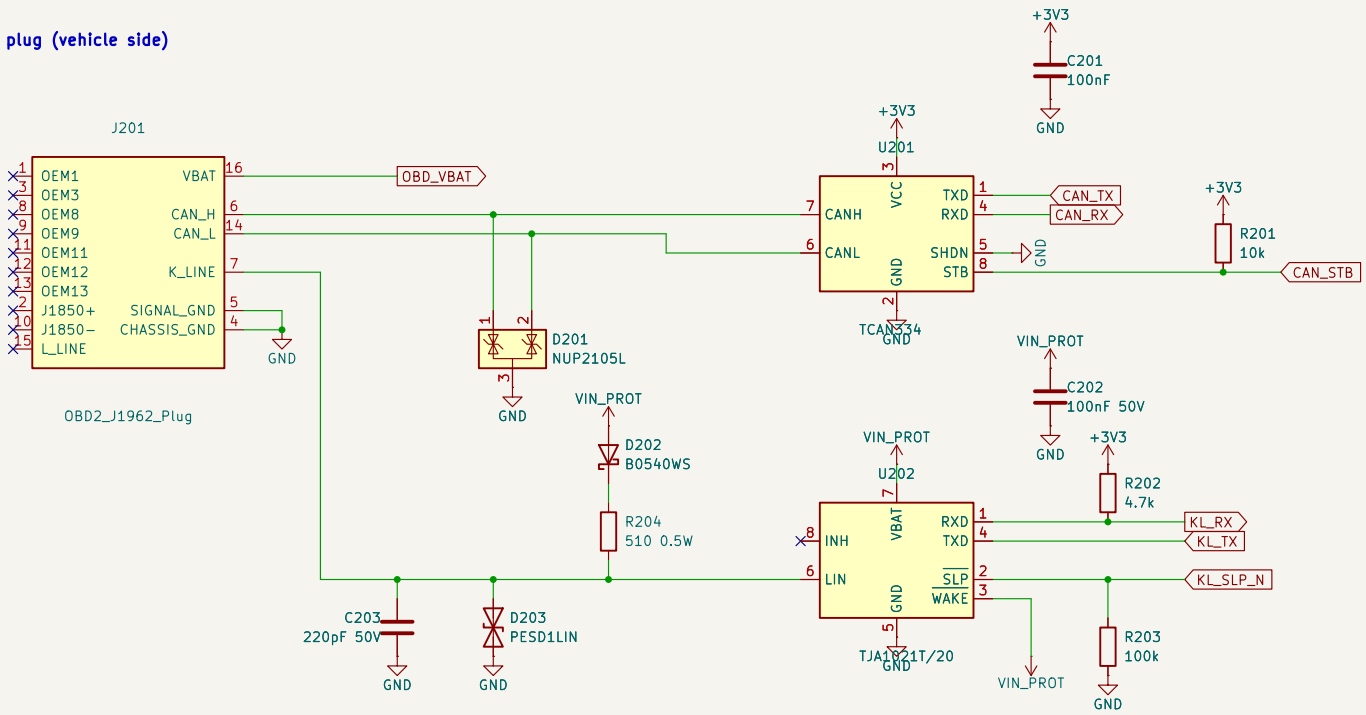
ERC power flags



Sheet: /Power/	
File: power.kicad_sch	
Title: Power: protection, buck 3.8 V, LDO 3.3 V	
Size: A4	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.1**
	Id: 2/6

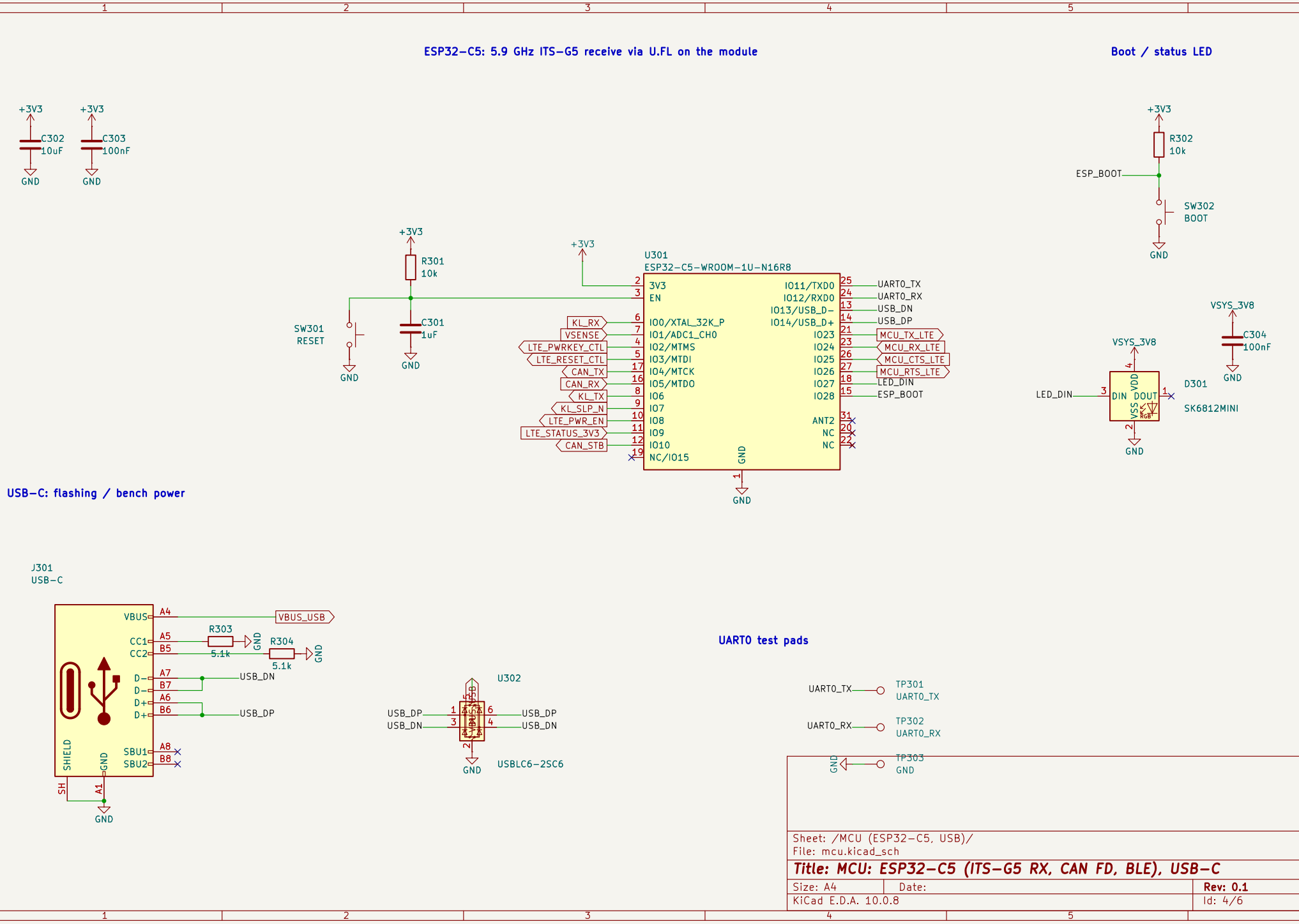
High-speed CAN (ISO 15765), no termination in the dongle

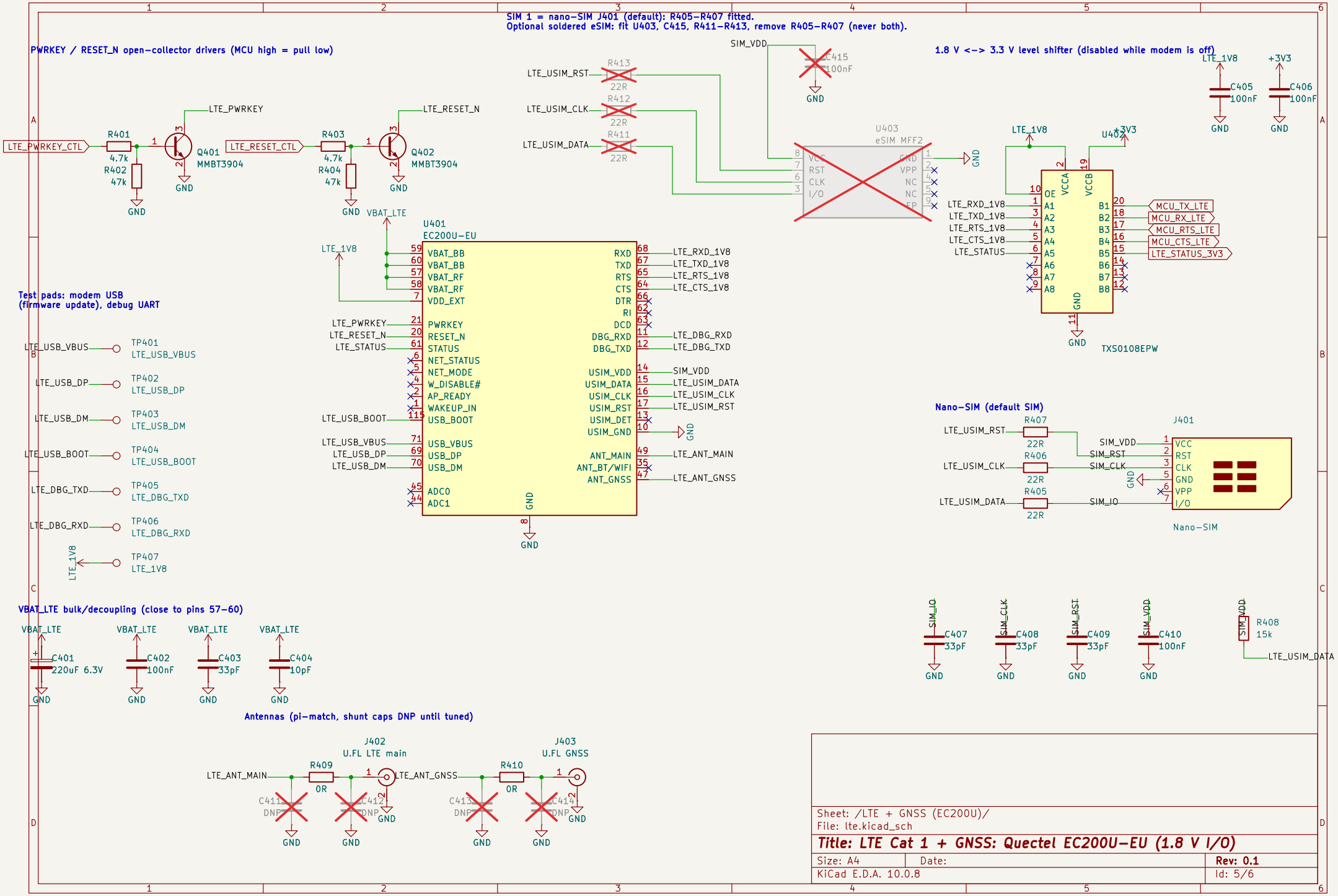
OBD-II plug (vehicle side)



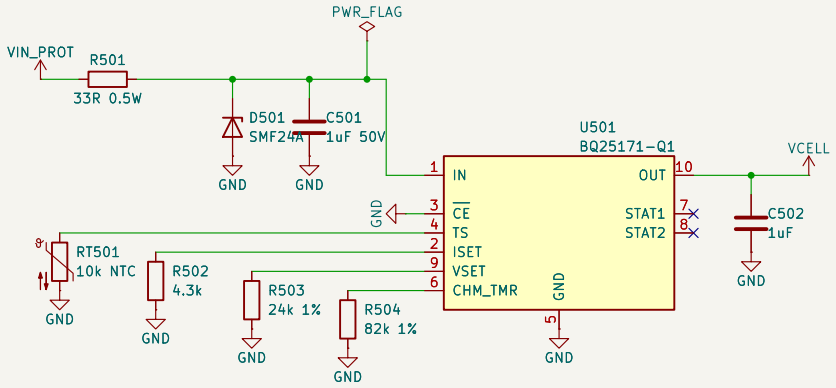
K-Line (ISO 9141 / KWP2000) via LIN transceiver + 510R tester pull-up

Sheet: /Vehicle (OBD, CAN, K-Line)/	
File: vehicle.kicad_sch	
Title: Vehicle interface: OBD-II, CAN, K-Line	
Size: A4	Date:
KiCad E.D.A. 10.0.8	Rev: 0.1
	Id: 3/6


ESP32-C5: 5.9 GHz ITS-G5 receive via U.FL on the module
Boot / status LED
USB-C: flashing / bench power
UART0 test pads
+3V3
C302
10uF
GND
C303
100nF
R301
10k
SW301
RESET
C301
1uF
U301
ESP32-C5-WROOM-1U-N16R8
3V3
EN
IO0/XTAL_32K_P
IO1/ADC1_CH0
IO2/MTMS
IO3/MTDI
IO4/MTCK
IO5/MTDO
IO6
IO7
IO8
IO9
IO10
NC/IO15
IO11/TXD0
IO12/RXD0
IO13/USB_D-
IO14/USB_D+
IO23
IO24
IO25
IO26
IO27
IO28
ANT2
NC
NC
KL_RX
VSENSE
LTE_PWRKEY_CTL
LTE_RESET_CTL
CAN_TX
CAN_RX
KL_TX
KL_SLP_N
LTE_PWR_EN
LTE_STATUS_3V3
CAN_STB
UART0_TX
UART0_RX
USB_DN
USB_DP
MCU_TX_LTE
MCU_RX_LTE
MCU_CTS_LTE
MCU_RTS_LTE
LED_DIN
ESP_BOOT
R302
10k
ESP_BOOT
SW302
BOOT
VSYS_3V8
C304
100nF
D301
SK6812MINI
J301
USB-C
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
SH
R303
5.1k
R304
5.1k
VBUS_USB
U302
USBLC6-2SC6
TP301
UART0_TX
TP302
UART0_RX
TP303
GND
Sheet: /MCU (ESP32-C5, USB)/
File: mcu.kicad_sch
Title: MCU: ESP32-C5 (ITS-G5 RX, CAN FD, BLE), USB-C
Size: A4
Date:
Rev: 0.1
KiCad E.D.A. 10.0.8
Id: 4/6

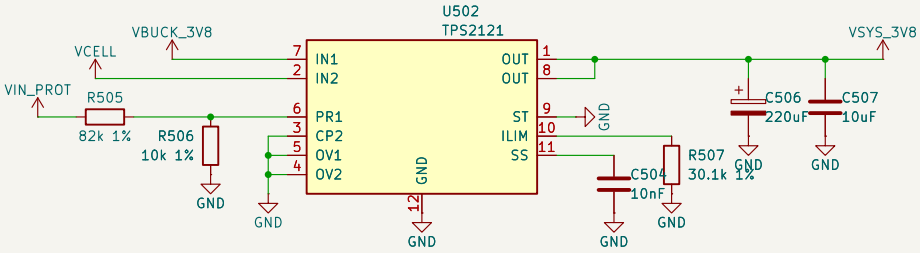

SIM 1 = nano-SIM J401 (default): R405-R407 fitted.
Optional soldered eSIM: fit U403, C415, R411-R413, remove R405-R407 (never both).
PWRKEY / RESET_N open-collector drivers (MCU high = pull low)
1.8 V <-> 3.3 V level shifter (disabled while modem is off)
Test pads: modem USB
(firmware update), debug UART
Nano-SIM (default SIM)
VBAT_LTE bulk/decoupling (close to pins 57-60)
Antennas (pi-match, shunt caps DNP until tuned)
U401
EC200U-EU
U403
eSIM MFF2
TXS0108EPW
Nano-SIM
J401
J402
U.FL LTE main
J403
U.FL GNSS
Sheet: /LTE + GNSS (EC200U)/
File: lte.kicad_sch
Title: LTE Cat 1 + GNSS: Quectel EC200U-EU (1.8 V I/O)
Size: A4
Date:
Rev: 0.1
KiCad E.D.A. 10.0.8
Id: 5/6

Li-ion charger, fed from the car side (VIN_PROT)



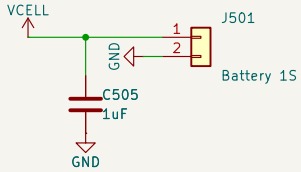
ISET 4.3k: 70 mA charge (kept low: linear from 14 V, ~0.6 W)
VSET 24k: 4.10 V float CHM_TMR 82k: Li-ion, 10 h timer
TS: 10k NTC next to the cell -> charging only 0..45 C

Priority power mux: buck (IN1) while VIN_PROT > ~9.7 V, else cell (IN2)



ILIM 30.1k: ~3.5 A SS 10 nF
Firmware on battery: LTE only (lower burst current), shut the modem down below ~3.6 V cell

1S Li-ion/LiPo cell, 250-400 mAh, with protection PCM



Sheet: /Backup battery/
File: battery.kicad_sch

Title: Backup battery: 1S Li-ion charger (BQ25171-Q1) + priority power mu

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 6/6